$$G = \nabla f(W)$$

$$X_0 = G/(\|G\|_F + \varepsilon)$$

$$\text{AOL: } P = 1.5I - 0.5X_0X_0^T$$

$$X_0 \leftarrow PX_0$$

repeat $k = 0 \ldots K' - 1$ ($K' \approx 4$):

$$A = X_kX_k^T$$

$$B = 1.5A - 0.5A^2$$

$$X_{k+1} = BX_k$$

$$W \leftarrow W - \eta X_K'$$